

VOLUME VI · CATALOGUE EXTENSION · ANSWERS "IS IT COMPLETE?"

Catalogue Extension

No — Volume III was not complete, and it was not trying to be. It covered soil fertility chemistry, which is what SoilFER needs. Going generic changed the requirement. Here are the 113 analyses that closes the gap, across five matrices the original catalogue never contemplated.

VOLUME III	THIS EXTENSION	CATALOGUE TOTAL
101 analyses · soil only	113 analyses · 5 new matrices	214 analyses · ~430 methodologies

STRUCTURAL CHANGE

Sample matrix becomes a first-class field

The honest answer is no, and it is worth being precise about why. Volume III's closing section said what it excluded: pesticide residues, radionuclides, XRD mineralogy and isotope work, on the grounds that "none belong in a fertility platform's default catalogue". That judgement was correct for SoilFER. Volume V then decided the platform ships generically to "any lab and institution", and that decision invalidated the judgement without anyone noticing.

The larger omission was not the exclusions, it was a whole axis. Volume III catalogued analyses of *soil*. A soil fertility laboratory routinely also analyses plant tissue, irrigation water, compost and manure, fertilizers, and liming materials — often on the same bench, for the same client, on the same report. Both project documents touch this directly: SoilFER Activity 3.3 commits to fertilizer quality analysis and to promoting "organic fertilizers, biological fertilizers, or the reuse and recycling of nutrients", and SoilFER-VACS Activity 2.5 commits to "Economic Viability and **Nutritional Analysis** of Opportunity Crops".

Neither is possible with a soil-only catalogue.

One structural consequence. Matrix cannot be a label — it changes what an analysis means. Total nitrogen in soil is 0.5–5 g/kg; in plant tissue 15–50 g/kg; in urea 460 g/kg. Same analyte, three plausible ranges, three interpretation frameworks, three sets of reference materials. Volume IV already flagged that a sample type field was needed for fertilizers. It is needed for all of it, and it must key the ranges, the bands and the QC.

21 PLANT TISSUE Diagnosis and crop nutritional analysis	19 WATER Irrigation suitability and salinity	18 AMENDMENTS Compost, manure, biochar, lime	17 FERTILIZERS Nutrient content and contaminants
27 SOIL, MISSED Health indicators, chemistry, biology, physics	11 ENVIRONMENTAL Opt-in contamination and mineralogy module		

GLOSOLAN	published	FAO SOP	ISO	international standard
REF	literature, EN or national method	DERIVED	calculated	recommended default

01 Matrix becomes a real field

The one schema change this extension forces, and the reason it cannot simply be appended to the existing tables.

Analysis	
code, name, categoryId, unitCode, decimalPlaces, resultType	
+ matrix	SOIL PLANT WATER AMENDMENT FERTILIZER LIMING OTHER
+ module	FERTILITY HEALTH ENVIRONMENTAL QUALITY – controls what a 1

Sample	
+ matrix	what physically arrived; constrains which analyses can be ordered
+ plantPart	leaf, grain, whole shoot – plant matrix only
+ growthStage	plant matrix only; interpretation depends on it entirely
ValidationRange	plausible min/max keyed on (analysis, matrix, method)
ThresholdSet	interpretation bands keyed on (analysis, matrix, method, crop, regi

Plant analysis makes the case on its own. A maize leaf sampled at silking and the same leaf sampled at maturity have different sufficiency ranges for every nutrient; a leaf and a grain differ more still. Without growth stage and plant part on the sample, a tissue result cannot be interpreted at all – which is why laboratories that bolt plant analysis onto a soil LIMS usually end up reporting numbers with no verdict, and the agronomist reads them off a printed table instead.

The `module` field is what keeps a small fertility laboratory from drowning. A lab that does soil and plant work should never see eleven contaminant analyses in its picker. Modules are enabled per laboratory in the profile; the environmental module ships disabled.

02

Plant tissue analysis

PLANT

The largest single omission. Tissue analysis is how a fertility laboratory diagnoses what soil analysis only predicts, and SoilFER-VACS commits to nutritional analysis of opportunity crops. GLOSOLAN has no plant SOPs – the reference frame here is AOAC and the standard handbooks, which is what plant laboratories actually cite.

ANALYSIS	UNIT	METHODS & REFERENCES
Dry matter / moisture PL_DM	% 1 dp	● Oven drying 65 °C then 105 °C AOAC 930.15
Ash content PL_ASH	% 2 dp	● Dry ashing at 550 °C AOAC 942.05
Total nitrogen PL_N	g/kg 2 dp · 5–60	● Dumas dry combustion AOAC 990.03 Kjeldahl digestion AOAC 978.04 · Kalra 1998
Crude protein PL_PROT	% 2 dp	● Total N × 6.25 (or crop-specific factor) DERIVED

Total phosphorus PL_P	g/kg 3 dp	● Dry ashing, ICP-OES A0AC 985.01 · Kalra 1998 - Wet digestion HNO₃/H₂O₂, colorimetric Kalra 1998
Total potassium PL_K	g/kg 2 dp	● Dry ashing, ICP-OES or flame photometry A0AC 985.01
Total calcium PL_CA	g/kg 2 dp	● Dry ashing, ICP-OES A0AC 985.01
Total magnesium PL_MG	g/kg 2 dp	● Dry ashing, ICP-OES A0AC 985.01
Total sulfur PL_S	g/kg 2 dp	● Dry combustion or wet digestion, ICP-OES Kalra 1998
Total iron PL_FE	mg/kg 1 dp	● Dry ashing, ICP-OES A0AC 985.01
Total manganese PL_MN	mg/kg 1 dp	● Dry ashing, ICP-OES A0AC 985.01
Total zinc PL_ZN	mg/kg 1 dp	● Dry ashing, ICP-OES A0AC 985.01
Total copper PL_CU	mg/kg 2 dp	● Dry ashing, ICP-OES A0AC 985.01
Total boron PL_B	mg/kg 2 dp	● Dry ashing, azomethine-H or ICP-OES Kalra 1998
Total molybdenum PL_MO	mg/kg 3 dp	● Wet digestion, ICP-MS Kalra 1998
Total sodium PL_NA	g/kg 2 dp	● Dry ashing, ICP-OES A0AC 985.01
Total chloride PL_CL	g/kg 2 dp	● Water extraction, argentometric or IC Kalra 1998
Nitrate in petiole / sap PL_N03	mg/kg 0 dp	● Water extraction, ion-selective electrode or IC Kalra 1998
Selenium PL_SE	mg/kg 3 dp	● Wet digestion, hydride generation AAS or ICP-MS A0AC 985.01
Heavy metals in tissue DI UM	mg/kg 3 dp	● Cd, Pb, As, Hg — microwave digestion, ICP-MS A0AC 2005.01

Crop

Nutrient sufficiency & ratios	index	● DRIS indices from nutrient concentrations DERIVED
PL_DRIS	2 dp	Sufficiency range comparison by crop and growth stage



03

Irrigation and agricultural water

WATER

Volume III catalogued salinity and sodicity on the soil side and stopped there. The cause of most salinity is the irrigation water, and a laboratory advising on the one is always asked about the other. FAO Irrigation and Drainage Paper 29 (Ayers & Westcot) is the interpretation framework the whole sector uses.

ANALYSIS	UNIT	METHODS & REFERENCES
pH W_PH	pH units 2 dp	● Electrometric ISO 10523:2008
Electrical conductivity W_EC	dS/m 3 dp	● Conductivity cell at 25 °C ISO 7888:1985
Total dissolved solids W_TDS	mg/L 0 dp	● Gravimetric at 180 °C APHA 2540 C From EC × factor DERIVED
Calcium W_CA	mmol(+) / L 2 dp	● ICP-OES ISO 11885:2007 EDTA titration APHA 3500-Ca
Magnesium W_MG	mmol(+) / L 2 dp	● ICP-OES ISO 11885:2007
Sodium W_NA	mmol(+) / L 2 dp	● ICP-OES or flame photometry ISO 11885:2007
Potassium W_K	mmol(+) / L 2 dp	● ICP-OES or flame photometry ISO 11885:2007
Carbonate & bicarbonate W_CO3	mmol(-) / L 2 dp	● Double-endpoint titration (alkalinity) ISO 9963-1:1994
Chloride W_CL	mmol(-) / L 2 dp	● Ion chromatography ISO 10304-1:2007 Argentometric (Mohr) APHA 4500-Cl⁻ B

Sulfate W_S04	mmol (-) / L 2 dp	● Ion chromatography ISO 10304-1:2007
Nitrate-N W_NO3	mg / L 2 dp	● Ion chromatography ISO 10304-1:2007
Ammonium-N W_NH4	mg / L 2 dp	● Spectrometric (indophenol blue) ISO 7150-1:1984
Phosphorus W_P	mg / L 3 dp	● Ammonium molybdate spectrometry ISO 6878:2004
Boron W_B	mg / L 3 dp	● ICP-OES ISO 11885:2007
Trace elements & heavy metals W_TRACE	mg / L 4 dp	● Fe, Mn, Zn, Cu, Cd, Pb, As, Ni — ICP-OES or ICP-MS ISO 11885 · ISO 17294-2
Sodium adsorption ratio W_SAR	(mmol / L) ^½ 2 dp	● Na ÷ √((Ca+Mg)/2) DERIVED Adjusted SAR (SAR _{adj}) accounting for carbonate precip FAO I&D Paper 29
Residual sodium carbonate W_RSC	mmol (-) / L 2 dp	● (CO₃ + HCO₃) – (Ca + Mg) DERIVED
Irrigation suitability class W_CLASS	class —	● Salinity, sodicity, toxicity and miscellaneous hazard FAO I&D Paper 29 (Ayers & Westcot)
Ion balance check W_BALANCE	% 1 dp	● Cation–anion balance, must close within ±5 % DERIVED

04

Organic amendments and liming materials

AMENDMENT

Both project documents commit to promoting organic and alternative nutrient sources — compost, manure, biological fertilizers, nutrient recycling. A laboratory cannot advise on them without analysing them, and the EN 13037–13040 family is the standard reference for soil improvers and growing media.

Sample preparation AM_PREP	—	● Sub-sampling and preparation of soil improvers
Moisture / dry matter AM_DM	% 1 dp	● Oven drying 105 °C EN 13040
pH AM_PH	pH units 1 dp	● Water suspension EN 13037
Electrical conductivity AM_EC	dS/m 2 dp	● Water extract EN 13038
Organic matter & ash AM_OM	% DM 1 dp	● Loss on ignition at 450–550 °C EN 13039
Organic carbon AM_OC	g/kg DM 1 dp	● Dry combustion ISO 10694:1995 From organic matter ÷ 1.8 DERIVED
Total nitrogen AM_N	g/kg DM 2 dp	● Dumas dry combustion ISO 13878:1998 Kjeldahl ISO 11261:1995
Ammonium-N and nitrate-N AM_MINN	mg/kg DM 1 dp	● KCl extraction — the plant-available fraction ISO
C:N ratio AM_CN	ratio 1 dp	● Organic C ÷ total N — the maturity indicator DERIVED
Total P, K, Ca, Mg, S AM_NUTR	g/kg DM 2 dp	● Aqua regia extraction, ICP-OES EN 13650
Micronutrients AM_MICRO	mg/kg DM 1 dp	● Fe, Mn, Zn, Cu, B, Mo — aqua regia, ICP-OES EN 13650
Heavy metals AM_HM	mg/kg DM 3 dp	● Cd, Cr, Cu, Hg, Ni, Pb, Zn — aqua regia, ICP-OES/M EN 13650 • ISO 11466
Respiration / stability AM_STAB	mg CO ₂ -C/g OM/d 2 dp	● Dynamic or static respiration index TMECC 05.08-B
Phytotoxicity / germination index AM_PHYTO	% 0 dp	● Seed germination and root elongation bioassay TMECC 05.08-B
Impurities and stones AM_IMPUR	% DM 2 dp	● Manual separation after sieving — glass, plastic, m EU Reg. 2019/1009 Annex IV

Neutralizing value LI_NV	% CaO eq 1 dp	● Acid dissolution and back-titration EN 12945
Liming material fineness LI_FINE	% passing 1 dp	● Dry sieving at defined apertures EN 12948
Effective calcium carbonate equivalent LI_ECCE	% 1 dp	● Neutralizing value × fineness efficiency factor DE

05

Fertilizer quality

FERTILIZER

Flagged in Volume IV, detailed here. SoilFER's founding argument is the fertilizer crisis and "the use of low-quality fertilizers that do not contain adequate proportion of nutrients and have high loads of contaminants" — both halves of that sentence need a catalogue entry.

ANALYSIS	UNIT	METHODS & REFERENCES
Moisture FE_MOIST	% 2 dp	● Karl Fischer or oven drying A0AC 950.01
Total nitrogen FE_N_TOT	% m/m 2 dp	● Kjeldahl after reduction A0AC 978.01 · EU Reg. 2019/
Ammoniacal nitrogen FE_N_NH4	% m/m 2 dp	● Distillation and titration A0AC 950.02
Nitric nitrogen FE_N_N03	% m/m 2 dp	● Reduction then distillation EU Reg. 2019/1009 Annex
Ureic nitrogen FE_N_UREA	% m/m 2 dp	● Urease hydrolysis then titration A0AC 976.01
Biuret FE_BIURET	% m/m 3 dp	● Colorimetric — the urea phytotoxicity contaminant
Total phosphorus (P ₂ O ₅) FE_P_TOT	% m/m 2 dp	● Mineral acid extraction, quinoline gravimetry or ICP
Water-soluble P ₂ O ₅	% m/m	● Water extraction — the immediately available fraction

02/09/2026, 10:28

FE_P_WS

2 dp

Catalogue Extension

A0AC 960.03

Neutral ammonium citrate-soluble P ₂ O ₅	% m/m 2 dp	• NAC extraction — with water-soluble gives "available"
FE_P_NAC		A0AC 960.03 • EU Reg. 2019/1009
Water-soluble potassium (K ₂ O)	% m/m 2 dp	• Water extraction, flame photometry or ICP-OES
FE_K		AOA
Sulfur	% m/m 2 dp	• Total and water-soluble SO ₃ , gravimetric or ICP
FE_S		EU Reg. 2019/1009 Annex IV
Calcium and magnesium	% m/m 2 dp	• Acid extraction, ICP-OES
FE_CAMG		EU Reg. 2019/1009 Annex IV
Micronutrients	% m/m 3 dp	• B, Co, Cu, Fe, Mn, Mo, Zn — total and water-soluble
FE_MICRO		EU Reg. 2019/1009 Annex IV
Heavy metals and contaminants	mg/kg 3 dp	• Cd, Pb, Hg, Ni, As, Cr(VI) in phosphate fertilizer mate
FE_HM		GLOSOLAN Heavy Metals in Phosphate Fertilizers 2024
Particle size / granulometry	% retained 1 dp	• Dry sieving — governs spreading uniformity
FE_PSD		EN 1235
Declared vs found compliance	% deviation 1 dp	• Measured against label declaration and permitted to
FE_COMPLY		DERIVED
Nutrient value per unit cost	currency/kg N 2 dp	• From measured content and purchase price
FE_VALUE		DERIVED

◀

▶

06

Soil analyses Volume III genuinely missed

SOIL

These are not matrix gaps — they are soil analyses that should have been in the original catalogue. The soil health group in particular is what most modern soil laboratories are being asked for, and SoilFER's own Output 1 is framed around soil health.

o

Soil health indicators

ANALYSIS	UNIT	METHODS & REFERENCES
Permanganate oxidizable carbon POXC	mg/kg 0 dp	0.02 M KMnO₄, colorimetric — "active carbon" Weil et al. 2018
Autoclaved citrate extractable protein ACE_PROT	mg/g 2 dp	Citrate extraction, autoclave, colorimetric — organic N Hurisso et al. 2018
Water-extractable organic carbon WEOC	mg/kg 1 dp	Cold or hot water extraction, TOC analyser Ghani et al. 2018
Water-extractable organic nitrogen WEON	mg/kg 1 dp	Same extract, total N minus mineral N DERIVED
Short-term respiration burst RESP_BURST	mg CO ₂ /kg 1 dp	24 h CO₂ after rewetting dried soil (Solvita or titration) Franzluebbers et al. 2000
Soil health index SH_INDEX	score 0 dp	Weighted scoring of biological, chemical and physical indicators DERIVED

P Additional soil chemistry

ANALYSIS	UNIT	METHODS & REFERENCES
Available silicon AVAIL_SI	mg/kg 1 dp	0.5 M acetic acid extraction, colorimetric Korndörfer et al. 2006 0.01 M CaCl₂ extraction Haysom & Chapman 1975
Reserve (non-exchangeable) potassium RESERVE_K	mg/kg 1 dp	Boiling 1 M HNO₃ Pratt 1965
Hydrolysable / potentially available N HYDROL_N	mg/kg 1 dp	Alkaline permanganate Subbiah & Asija 1956
Phosphorus fractionation P_FRACT	mg/kg 2 dp	Sequential extraction — labile, Fe/Al-bound, Ca-bound Hedley et al. 1982

Available selenium AVAIL_SE	mg/kg 3 dp	● Phosphate or KH ₂ PO ₄ extraction, HG-AAS or ICP-MS
Available iodine AVAIL_I	mg/kg 3 dp	● Alkaline extraction, ICP-MS — biofortification relevance Watts & Mitchell 2009
Available cobalt AVAIL_CO	mg/kg 3 dp	● DTPA or acetic acid extraction ISO 14870:2001
Available nickel AVAIL_NI	mg/kg 3 dp	● DTPA extraction GLOSOLAN DTPA 2023
Water-soluble chloride SOIL_CL	mg/kg 1 dp	● 1:5 water extraction, IC or argentometric ISO 10304-1:2001
Water-soluble fluoride SOIL_F	mg/kg 2 dp	● Water extraction, ion-selective electrode ISO 10359-1:2001
Redox potential REDOX	mV 0 dp	● Platinum electrode — essential for paddy and wetland Patrick et al. 1996
Total aluminium TOTAL_AL	g/kg 2 dp	● HF or aqua regia digestion, ICP-OES ISO 14869-1:2001
Ammonium fixation capacity NH4_FIX	mg/kg 1 dp	● Fixed NH ₄ after HF-HCl treatment Silva & Bremner 1990
Salinity class SAL_CLASS	class —	● Saline / sodic / saline-sodic from EC, ESP and pH DERIVED



q Soil biology extensions

ANALYSIS	UNIT	METHODS & REFERENCES
Soil DNA extraction BIO_DNA	µg/g 2 dp	● Direct extraction of soil DNA ISO 11063:2020
Microbial abundance by qPCR BIO_QPCR	copies/g —	● 16S / ITS gene copy quantification Lab SOP · ISO 11063:2020

Microbial community (PLFA) BIO_PLFA	nmol/g 2 dp	● Phospholipid fatty acid profiling, GC Frostegård et al
Earthworm abundance BIO_WORM	count/m² 0 dp	● Hand-sorting and extraction ISO 23611-1:2018
Soil macro-invertebrates BIO_MACRO	count/m² 0 dp	● Sampling and extraction of macro-invertebrates ISO
Micro-arthropods BIO_ARTH	count/sample 0 dp	● Extraction of micro-arthropods (Collembola, Acari) ISO 23611-2:2006
Mycorrhizal root colonization BIO_AMF	% 1 dp	● Root clearing, staining and gridline intersect Giovannetti & Mosse 1980



R Additional soil physics

ANALYSIS	UNIT	METHODS & REFERENCES
Dispersion / slaking (Emerson) EMERSON	class 1–8 —	● Emerson aggregate coherence test Emerson 1967
Modulus of rupture / crusting MOD RUPT	kPa 1 dp	● Briquette breaking strength USDA SSIR 42
Air permeability AIR_PERM	µm² 2 dp	● Steady-state airflow through an undisturbed core Ball & Schjønning 2002
Soil temperature SOIL_TEMP	°C 1 dp	● In-situ probe at stated depth Field record
Field water content at sampling FIELD_WC	% w/w 1 dp	● Gravimetric on field-moist sample — needed for mineral ISO 11465:1993



07

Environmental module — ships disabled

ENVIRONMENTAL

Volume III excluded these deliberately and, for SoilFER, correctly. For a generic product they belong in the catalogue but behind a switch: a fertility laboratory should never see them, and a contamination laboratory should not have to fork the repository to add them.

ANALYSIS	UNIT	METHODS & REFERENCES
Elemental composition by XRF ENV_XRF	mg/kg 1 dp	● Wavelength or energy dispersive XRF, non-destructive ISO 18227:2014
Hydrocarbon content C10–C40 ENV_TPH	mg/kg 1 dp	● Solvent extraction, GC-FID ISO 16703:2004
Polycyclic aromatic hydrocarbons ENV_PAH	mg/kg 3 dp	● GC-MS, 16 EPA priority PAHs ISO 18287:2006
Organochlorine pesticides & PCBs ENV_OCP	µg/kg 2 dp	● GC with electron capture detection ISO 10382:2002
Multi-residue pesticide screen ENV_PEST	µg/kg 2 dp	● QuEChERS extraction, LC-MS/MS and GC-MS/MS
Glyphosate and AMPA ENV_GLYPH	µg/kg 2 dp	● Derivatisation, LC-MS/MS Lab SOP
Leaching behaviour ENV_LEACH	mg/kg 3 dp	● Batch leaching at fixed liquid-to-solid ratio ISO 21468
Radionuclide activity ENV_RAD	Bq/kg 2 dp	● Gamma spectrometry — K-40, Cs-137, Ra-226, Th-232, U-235 IAEA TRS-295
Clay mineralogy ENV_XRD	% / semi-quant 1 dp	● X-ray diffraction on oriented clay fraction Whittig & Allardice 1986
Gamma-ray spectrometry scan SPEC_GRS	counts / Bq/kg 2 dp	● In-situ or laboratory GRS — K, U, Th windows IAEA 10543
Properties estimated from GPR	as analyte	● Texture and mineralogy from radioelement ratios